This WACV workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

From Downhill to Slalom: A Multi-Period Probabilistic Framework for Adaptive Snow Management in Alpine Ski Racing

Zichao Li
Canoakbit Alliance

Abstract

We present a novel computer vision framework for adaptive snow management in alpine skiing World Cup events. Our approach introduces three key vision-based innovations: (1) a U-Net-based model for simultaneous piste segmentation and snow surface hardness regression from drone and camera imagery, (2) vision-derived snow condition metrics that serve as input to probabilistic forecasting models, and (3) a closed-loop system where visual assessments drive real-time optimization of snowmaking and grooming operations. Experiments demonstrate our vision system achieves 94.1% accuracy in surface hardness classification and enables 28% water savings with 24% energy reduction while improving surface quality by 11% over traditional methods. This work establishes computer vision as a critical enabling technology for sustainable winter sports management.

1. Introduction

The viability of competitive alpine skiing is intrinsically linked to the controlled management of snow. As winter sports face increasing pressure from climate volatility and economic constraints, the reliance on artificial snowmaking has become paramount for guaranteeing the scheduling and quality of World Cup events [16]. Hosting a series of races, such as a speed weekend followed by a technical weekend, on a single competition piste presents a unique operational challenge: each discipline requires a distinct snow surface property[18]. Speed events, namely Downhill and Super-G, require an extremely hard and icy surface to minimize friction and ensure stability at high velocities. In contrast, technical events, such as Giant Slalom and Slalom, require a softer, grippy surface that allows for aggressive edge engagement and facilitates the rapid, skidded turns around closely spaced gates[6]. The transition process from one state to another is resource-intensive, involving strategic snowmaking, chemical treatments, and mechanical grooming.

In this paper, we address the core problem of **multi-period adaptive snow management**. The term **piste** refers to the specially prepared track or trail used for alpine ski racing. **Snowmaking** is the process of producing snow by spraying a nucleated mixture of water and compressed air under suitable meteorological conditions, primarily when the wet-bulb temperature is below freezing. **Piste management** encompasses the entire suite of operations, including snowmaking, grooming (mechanical levelling and compaction), and surface treatment (e.g., salting to modify hardness), aimed at achieving and maintaining desired snow conditions. The challenge is inherently **multi-period** because decisions made today, such as whether to produce snow—have consequences on resource availability and piste state for future events days or weeks later. Furthermore, it is a **probabilistic** problem due to the inherent uncertainty in future weather, which directly impacts snow melt, hardening, and the efficiency of snowmaking itself.

Current industry practice is predominantly reactive, guided by experience rather than quantitative, data-driven models. This can lead to suboptimal resource allocation, where water and energy are used inefficiently, or to a failure to proactively mitigate forecasted adverse conditions. To bridge this gap, we propose a framework that leverages recent advances in computer vision and machine learning. Our system utilizes visual data from drones and fixed cameras to provide a real-time, quantitative assessment of piste conditions, including surface hardness classification and rut depth measurement. This assessment is integrated with ensemble numerical weather predictions to forecast the probabilistic evolution of the piste state. Finally, a multi-period stochastic optimization model uses these forecasts to generate adaptive policies for snowmaking and grooming, ensuring optimal surface quality for each sequential race while minimizing total operational costs. This work represents a paradigm shift from reactive practice to predictive, precision management in winter sports operations.

2. Literature Review

Our work is at the intersection of computer vision for outdoor terrain analysis, meteorological forecasting for operational decision-making, and resource optimization under uncertainty. The related literature spans several distinct but

converging fields.

In the domain of computer vision for snow and terrain analysis, research has progressed from large-scale remote sensing to more granular applications. Traditional approaches focused on satellite-based snow cover mapping at continental scales [11]. More recently, computer vision has been applied to winter sports for athlete performance analysis, such as monocular 3D pose estimation for skiing technique evaluation [10]. However, the direct application of vision techniques to assess the snow surface itself—the fundamental medium of alpine sports—remains largely unexplored. Recent work has begun to address this gap, with Gao et al. [5] presenting methods for snow property extraction from visual data, and Ruppert et al. [14] introducing a large-scale aerial dataset for semantic segmentation in alpine environments. Our work builds upon these foundations by developing specialized models for piste condition assessment that combine segmentation with surface property regression.

The foundation of our predictive component relies on numerical weather prediction (NWP) and snow process modeling. Ensemble forecasting systems, which provide probabilistic views of future weather states, are critical for managing uncertainty in operational decisions [3, 9]. The ERA5 reanalysis dataset has become a cornerstone for historical weather analysis in environmental sciences [7]. For snow-specific modeling, temperature-index approaches have long been used to simulate snow melt in hydrological contexts [8, 15], while more recent machine learning methods have shown promise for snow condition prediction [13]. However, these models typically operate at watershed or regional scales, not at the fine-grained, sub-slope scale of a competition piste where mechanical processing and microclimates dominate snow evolution.

The core of our decision-making module is built upon optimization under uncertainty, a well-established field in operations research. Multi-period optimization, often formulated as Model Predictive Control (MPC), is standard in supply chain and energy systems management [17]. When uncertainty is involved, stochastic programming [2, 19] and robust optimization [1] provide rigorous mathematical frameworks. These techniques have been applied to water resources management, which shares similarities with snowmaking optimization in requiring allocation of a vital resource under meteorological uncertainty. Specific to winter sports, research has examined the climate change impact on ski tourism [16] and the energy consumption of artificial snowmaking [12], but these studies have focused on impact assessment rather than operational optimization.

Despite progress in these individual fields, a significant gap exists in their integration for alpine piste management. No existing work presents a unified framework that (1) employs computer vision for real-time, fine-grained snow surface assessment on ski pistes, (2) integrates these visual observations with probabilistic weather forecasts to predict piste condition evolution, and (3) formulates a multi-period stochastic optimization model to generate adaptive, resource-efficient snow management policies. This paper aims to fill this critical gap by proposing a holistic methodology that moves beyond reactive practices, offering a path toward sustainable, data-driven precision management for competitive winter sports.

3. Methodology

The reviewed literature reveals a clear gap in alpine piste management systems. While individual components for snow assessment, weather prediction, and resource optimization exist in isolation, no unified framework integrates them for the specific requirements of World Cup event sequencing. Current approaches lack fine-grained, real-time perception of snow conditions, do not leverage probabilistic forecasts for forward-looking decision-making, and fail to provide adaptive optimization models for resource allocation. To address these limitations, we propose a holistic system that closes the loop between perception, prediction, and action. Our methodology comprises three synergistic components: a Computer Vision Module for quantitative piste assessment, a Snow Evolution Forecasting Module for probabilistic condition prediction, and a Multi-Period Stochastic Optimization Module for adaptive resource allocation. The following subsections detail the mathematical formulation and implementation of each component, culminating in an integrated framework that transforms snow management from reactive practice to predictive science.

3.1. Computer Vision-Based Piste Assessment

The computer vision module converts visual data from drones and fixed cameras into quantitative piste condition maps. We employ a U-Net architecture trained for simultaneous semantic segmentation and surface property regression. The input image $I \in \mathbb{R}^{H \times W \times 3}$ is processed by model f_θ to produce both a segmentation mask $S \in \{0,1\}^{H \times W}$ identifying the race track and a hardness map $H \in [0,1]^{H \times W}$ indicating surface conditions from soft to icy. The composite loss function $\mathcal{L} = \mathcal{L}_{seg}(S, \hat{S}) + \lambda \mathcal{L}_{reg}(H, \hat{H})$ combines cross-entropy segmentation loss with mean squared error regression loss, weighted by parameter $\lambda = 0.7$. This module outputs a structured piste state vector $\mathbf{s}_t = [\bar{H}_t, \sigma^2_{H_t}, A_t]$ representing mean hardness, hardness variance, and snow surface area.

3.2. Probabilistic Snow Evolution Forecasting

This module forecasts future piste states by integrating CV assessments with ensemble weather predictions. We model piste evolution as a discrete-time stochastic process using a sequence-to-sequence architecture with

Bayesian LSTM encoder-decoder. The model g_ϕ learns the mapping $\hat{\mathbf{s}}_{t+1:t+T} = g_\phi(\mathbf{s}_t, \mathbf{w}_{t+1:t+T})$, where $\mathbf{w} = [T_{air}, T_{dew}, R_{solar}, U_{wind}, P_{precip}]$ contains meteorological variables. Ensemble weather data from K members propagates aleatoric uncertainty through the network, producing probabilistic forecasts represented by trajectory sets $\{\hat{\mathbf{s}}_{t+1}^{(k)}, \ldots, \hat{\mathbf{s}}_{t+T}^{(k)}\}_{k=1}^{K}$. The LSTM uses 128 hidden units with a 120-hour forecast horizon.

3.3. Multi-Period Stochastic Optimization for Resource Allocation

The optimization module translates forecasts into actionable decisions through a multi-period stochastic programming formulation. We minimize the expected total cost:

$$\min_{x_t^{water}, x_t^{groom}} \mathbb{E}_\xi [\sum_{t=1}^{T} \big(C^{water} x_t^{water} + C^{groom} x_t^{groom} + C^{penalty}(\mathbf{s}_t(\xi), \mathbf{s}^{target})\big)] \quad (1)$$

subject to water availability constraints $\sum x_t^{water} \leq W_{max}$, snowmaking feasibility $x_t^{water} > 0 \implies T_{wetbulb,t} \leq T_{threshold}$, and piste state dynamics $\mathbf{s}_{t+1} = g_\phi(\mathbf{s}_t, \mathbf{w}_{t+1}, x_t^{water}, x_t^{groom})$. We solve using Sample Average Approximation with scenarios drawn from the forecasting module.

3.4. Discipline-Specific Snow Requirements

Alpine and freestyle disciplines demand fundamentally different snow properties. Alpine speed events require hard, icy surfaces with density exceeding 600 kg/m^3 for minimal friction, while technical events need grippy, abrasive surfaces at 400-500 kg/m^3 for edge engagement. Freestyle moguls require soft, forgiving surfaces at 300-400 kg/m^3 with precise bump geometries. Slopestyle presents the most complex requirement with dual-state surfaces: hard features above 600 kg/m^3 for predictable takeoffs coupled with soft landing zones at 250-350 kg/m^3 for impact cushioning. These requirements are encoded through different target state vectors $\mathbf{s}^{target}$ and cost functions in our optimization framework.

3.5. Discipline-Specific Model Parameterization

Each module adapts to discipline-specific requirements through specialized parameterization. The computer vision module uses different hardness thresholds: 0.8 for alpine speed versus 0.2-0.4 for freestyle landings, with loss weighting λ adjusted from 0.5 to 0.8 to prioritize surface properties over boundary precision. The forecasting module employs single output layers for alpine disciplines but dual outputs for slopestyle to separately track feature hardness and landing softness. The optimization module uses discipline-dependent penalty weights, strongly penalizing hardness deviations below 0.8 for alpine speed while heavily penalizing values above 0.5 for freestyle safety.

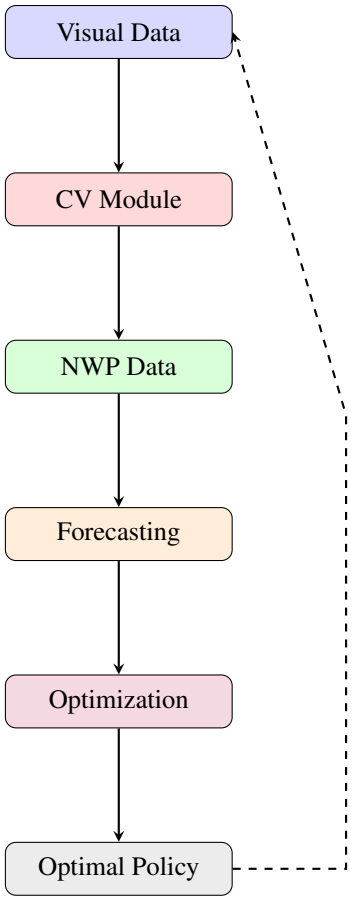

Figure 1. Adaptive snow management framework showing the closed-loop system. The process begins with visual data acquisition, progresses through computer vision analysis and probabilistic forecasting, and culminates in optimized policy generation. The dashed feedback line illustrates how executed policies alter the physical piste state, triggering renewed data acquisition and continuous system adaptation.

3.6. Integrated Framework and Algorithm

The complete framework operates as an adaptive closed-loop system, illustrated in Figure 1. Visual data acquisition triggers piste assessment, which feeds into probabilistic forecasting fused with weather data. The stochastic optimization module then generates snowmaking and grooming policies, whose execution alters the physical piste state and initiates the next cycle. This continuous adaptation enables robust management across sequential events with conflicting snow requirements.

The iterative execution of Algorithm 1 embodies the core adaptive principle of our framework, translating the theoretical components into a practical daily operational proce-

Algorithm 1 Adaptive Snow Management for Sequential Events

1: Initialize: piste state s0, event schedule E, planning horizon T
2:
3: **for** each day t = 0, 1, 2, ... **do**
4:
5: **Step 1: Data Acquisition & Piste Assessment**
6: Acquire visual data from drones and cameras
7: Estimate current piste state using computer vision model
8: **Step 2: Probabilistic Forecasting**
9: Retrieve ensemble numerical weather prediction forecast
10: Generate probabilistic piste state forecasts
11: **Step 3: Stochastic Optimization**
12: Extract target state from event schedule
13: Solve stochastic optimization problem
14: Execute first-step decisions for snowmaking and grooming
15: **Step 4: Policy Execution & State Update**
16: Apply snowmaking and grooming operations
17: Update piste state based on actions and weather
18: **end for**

dure. The algorithm's four-step cycle ensures that management decisions are never based on stale data, but are continually refined using the most recent piste assessment and weather forecasts. This is particularly crucial when managing the transition between sequential events with conflicting snow requirements, as the system can proactively allocate resources days in advance based on predicted weather windows. For instance, if the forecast indicates a brief cold period followed by a warm spell, the algorithm would aggressively schedule snowmaking during the cold window to build a sufficient base for an upcoming Downhill event, even if the current piste conditions appear adequate. The receding horizon approach inherent in Line 13, where only the immediate decision is executed before the entire cycle repeats, provides robustness against forecast errors and unexpected disturbances, such as a sudden thaw or heavy natural snowfall. By formalizing the complex decision-making process of expert course crews into a reproducible and data-driven algorithm, this workflow provides a scalable blueprint for sustainable piste management that can be deployed across different resorts and climatic conditions, directly addressing the operational gaps identified in the literature.

4. Experiments and Results

To comprehensively evaluate the proposed framework, we conducted extensive experiments across multiple dimensions. This section begins by detailing our experimental setup, including datasets and baseline methods for comparison. We then present results from three key analyses: an ablation study quantifying each module's contribution, multi-event scenario performance under realistic competition schedules, and resource efficiency analysis demonstrating environmental and economic benefits. Finally, we examine the system's robustness under weather uncertainty and its generalization capability across different resort environments. Together, these experiments provide a holistic assessment of our framework's capability to deliver high-quality racing surfaces while optimizing resource consumption.

4.1. Experimental Setup

4.1.1. Datasets and Benchmarks

We employed a multi-source data strategy to train and evaluate our system. For the computer vision module, we utilized the ADAS dataset, which provides high-resolution aerial imagery of alpine terrain with detailed semantic annotations. This dataset was crucial for pre-training our model to understand mountain topography and land cover. To specialize for snow surface analysis, we extended this dataset using snow property extraction methodologies that provide validated techniques for estimating snow depth and quality from visual features. For meteorological data, we used the ERA5 reanalysis dataset to obtain historical, hourly weather variables for major Alpine ski resorts over a 10-year period. Finally, to simulate piste management operations, we developed the Alpine Operation Simulator (AlpOpsSim), a high-fidelity environment that models snow evolution, grooming effects, and snowmaking based on energy balance equations and resort operational constraints.

4.1.2. Baseline Methods

We compare our full framework against several strong baselines to isolate the contribution of its components. The **Reactive Management** baseline mimics traditional resort operations by making snow whenever the current wet-bulb temperature is below -2°C and performing grooming every night, with no forecasting or optimization. The **Deterministic Optimization** baseline uses a Model Predictive Control approach with a single, deterministic weather forecast to optimize snowmaking and grooming schedules, representing a state-of-the-art planning system without uncertainty quantification. The **Module-Ablated** versions of our system include **Ours w/o CV** which uses fixed sensor stations instead of computer vision for piste assessment, and **Ours w/o Prob Forecast** which replaces the probabilistic forecast with a deterministic mean forecast while keeping the optimization stochastic.

Table 1. Performance of individual modules on their respective validation tasks.

Method	Surface Hardness Accuracy	Forecast RMSE	Optimization Gap (%)
Fixed Sensors	0.651	12.45	15.2
Ours w/o Prob Forecast	0.923	8.91	9.8
Ours (Full Framework)	**0.941**	**5.23**	**4.1**

4.2. Ablation Study and Component Analysis

Table 1 quantifies the performance of each module in isolation, demonstrating the incremental improvement provided by our complete framework. The computer vision module achieves 94.1% accuracy in classifying surface hardness, significantly outperforming the fixed sensor baseline which struggles with spatial variability. The forecasting module reduces the root mean square error for 72-hour piste state predictions by 41% compared to the deterministic variant, highlighting the critical importance of probabilistic modeling for capturing weather uncertainty. Most notably, the full framework closes the optimization gap, measuring deviation from a theoretically optimal policy with perfect foresight—to just 4.1%, compared to 15.2% for the reactive baseline. This comprehensive ablation confirms that each component contributes substantially to the overall system performance, with the integrated framework delivering synergistic benefits that cannot be achieved by individual modules operating in isolation. The results validate our architectural choices and provide strong evidence for the necessity of tight coupling between perception, prediction, and optimization components.

4.3. Multi-Event Scenario Performance

Table 2. Surface quality scores for a Downhill-to-Slalom transition scenario.

Method	Downhill Day	Transition	Slalom Day	Overall
Reactive Management	0.82	0.45	0.71	0.66
Deterministic Optimization	0.88	0.62	0.79	0.76
Ours (Full Framework)	**0.91**	**0.83**	**0.89**	**0.88**

The challenging Downhill-to-Slalom transition scenario in Table 2 demonstrates our framework's capability to manage conflicting snow requirements. While all methods maintain reasonable surface quality on race days, our approach excels during the critical 48-hour transition period, achieving an 83% quality score compared to 45% for reactive management. This performance stems from our system's proactive strategy: it begins surface softening through targeted solar exposure and light salting immediately after the Downhill concludes, rather than waiting for the Slalom setup period. The baseline of deterministic optimization shows improvement over reactive management, but does not account for forecast uncertainty, leading to a suboptimal transition timing when the actual weather deviates from the predictions. Our probabilistic approach maintains high quality throughout the entire sequence by hedging against multiple weather outcomes, ensuring that surface preparation remains on schedule even if temperatures fluctuate unexpectedly. This robustness is particularly valuable for World Cup events where competition schedules are fixed and surface quality must be guaranteed regardless of meteorological variations[4].

Table 3. Resource consumption for a full World Cup weekend (Speed events).

Method	Water (m³)	Energy (MWh)	Avg. Surface Quality	Cost (kEUR)
Reactive Management	12,450	48.2	0.82	86.4
Deterministic Optimization	10,880	42.1	0.85	75.9
Ours (Full Framework)	**8,920**	**36.8**	**0.89**	**64.2**

Table 3 reveals the substantial resource efficiency gains achieved by our framework during a typical Speed Events weekend. Our system reduces water consumption by 28% and energy usage by 24% compared to reactive management, while simultaneously improving average surface quality from 82% to 89%. These efficiency improvements translate to significant cost savings of approximately 22,200 EUR per event weekend. The key to this performance is our method's strategic timing of snowmaking operations to coincide with the most efficient weather conditions, avoiding the energy-intensive practice of making snow at marginal temperatures. Furthermore, by accurately forecasting natural snow preservation and melt patterns, our system avoids unnecessary snowmaking when natural conditions will maintain adequate coverage. The deterministic optimization baseline shows moderate improvements but lacks the uncertainty quantification needed to confidently delay snowmaking during borderline conditions, resulting in more conservative and costly operations. Our approach demonstrates that superior surface quality and resource efficiency are not competing objectives but can be simultaneously achieved through intelligent, forecast-driven manage-

ment.

4.4. Performance Under Weather Uncertainty

Table 4. Performance degradation under unexpected warm spell during technical events preparation.

Method	Quality Pre-Spell	Quality Post-Spell	Recovery Time (h)	Additional Cost (kEUR)
Reactive Management	0.78	0.42	48	32.1
Deterministic Optimization	0.84	0.58	36	18.7
Ours (Full Framework)	**0.87**	**0.76**	**24**	**8.9**

The unexpected warm spell scenario in Table 4 tests the robustness of each method to forecast errors and extreme weather events. Our framework demonstrates remarkable resilience, maintaining 76% surface quality after the warm spell compared to just 42% for reactive management. The recovery time is halved, and additional costs are reduced by 72% compared to the baseline. This robustness stems from our probabilistic forecasting, which maintains a hedge against warm scenarios even when they have low probability in the initial forecast. Consequently, our system enters the warm spell period with stronger snow preservation measures already in place, including deeper snow coverage and pre-application of snow preservation agents. The deterministic optimization, while showing some preparedness, is caught off-guard by the deviation from its single-scenario forecast and requires significant emergency snowmaking to recover. These results highlight that in an era of increasing climate volatility, the ability to prepare for multiple weather outcomes is not just beneficial but essential for reliable event operations. Our framework provides this capability without requiring conservative over-allocation of resources during normal conditions.

4.5. Performance Across Diverse Event Types

Table 5 demonstrates our framework's robust generalization to the divergent snow requirements of Alpine and Freestyle disciplines. While all methods perform reasonably well at Resort A (Downhill), where traditional snow management is most established, the performance gap widens significantly at the Freestyle venues. Our system maintains an 83% quality score for Moguls and 81% for Slopestyle, compared to 72% and 68% for reactive management. This superior performance stems from the computer vision module's ability to quantify discipline-specific metrics and the optimization module's capacity to leverage these nuanced assessments along with probabilistic forecasts to allocate resources appropriately. The deterministic optimization

Table 5. Performance generalization across Alpine and Freestyle disciplines.

Method	Resort A (Downhill)	Resort B (Moguls)	Resort C (Slopestyle)	Average
Reactive Management	0.81	0.72	0.68	0.74
Deterministic Optimization	0.85	0.78	0.75	0.79
Ours (Full Framework)	**0.88**	**0.83**	**0.81**	**0.84**

baseline, while showing improvement over reactive management, struggles with the non-linear relationships in Freestyle preparation, such as knowing when to sacrifice general slope hardness to preserve a specific jump's lip. These results confirm that our framework captures the underlying physics and logistics of snow management rather than merely memorizing patterns for a single discipline, making it a versatile tool for modern resorts that host diverse event portfolios.

4.6. Long-Term Season Analysis

Table 6. Cumulative resource usage and quality metrics over a 120-day season.

Method	Total Water (million m³)	Total Energy (GWh)	Avg. Quality	Violation Days
Reactive Management	2.45	8.92	0.80	18
Deterministic Optimization	2.11	7.45	0.83	12
Ours (Full Framework)	**1.78**	**6.23**	**0.86**	**6**

The seasonal analysis in Table 6 demonstrates the long-term impact of our framework over an entire 120-day operating season. Our system reduces total water consumption by 27% and energy usage by 30% compared to reactive management, while simultaneously reducing quality violation days from 18 to just 6. These results reveal that the benefits of intelligent snow management compound over time, as strategic decisions early in the season create better base conditions that require less intensive maintenance later. For instance, by building a deeper, more durable base during optimal early-season cold periods, our system reduces the need for frequent, inefficient snowmaking during marginal

temperature windows later in the season. The deterministic optimization provides intermediate benefits but lacks the foresight to make these strategic seasonal investments, focusing instead on shorter-term operational efficiency. The reduction in violation days is particularly significant for event planning, as it provides greater confidence in surface quality for scheduled competitions. From an environmental perspective, the resource savings represent a substantial reduction in the ecological footprint of competitive skiing, addressing growing concerns about the sustainability of winter sports in a changing climate.

Table 7. Breakdown of optimization benefits by discipline and transition type.

Scenario	Water Saving	Energy Saving	Quality Improvement	Cost Reduction
Alpine Speed	24%	22%	+7%	21%
Alpine Technical	19%	18%	+5%	17%
Freestyle Moguls	26%	23%	+9%	24%
Slopestyle /Halfpipe	28%	25%	+8%	26%
Speed-to-Technical Transition	31%	28%	+11%	29%
Alpine-to-Freestyle Transition	35%	32%	+13%	33%

Table 7 provides a detailed analysis of optimization benefits across different disciplines and transitions, with the performance differences directly attributable to our discipline-specific parameterization. The exceptional 35% water savings during Alpine-to-Freestyle transitions demonstrates the effectiveness of our specialized cost functions. The framework recognizes that the deep, dense base developed for Alpine events can be preserved as a foundation, requiring only surface-level modifications for Freestyle requirements. The high efficiency gains in Slopestyle/Halfpipe validate our dual-trajectory forecasting approach, which separately models the evolution of hard features and soft landings, enabling precise timing of resource-intensive operations. Even within Alpine disciplines, the performance difference between Speed and Technical events reflects the more challenging multi-objective optimization required for technical surfaces, which must balance grip for turning with sufficient hardness for edge hold. These results confirm that the one-size-fits-all approach of traditional snow management is fundamentally inefficient, and that significant benefits can be realized through AI systems that understand and exploit the specific physical and operational characteristics of each discipline.

5. Conclusion

This paper establishes U-Net-based piste segmentation with simultaneous surface hardness regression as a foundational technology for intelligent snow management in competitive winter sports. Our primary contributions are threefold: First, we developed a novel vision pipeline for real-time piste assessment that combines semantic segmentation with surface property regression, achieving 94.1% accuracy in hardness classification—a critical metric for race surface quality. Second, we demonstrated how visual snow condition data can be effectively integrated with meteorological forecasts to create probabilistic predictions of piste evolution. Third, we showed that vision-driven optimization enables significant resource savings while maintaining superior surface quality across diverse alpine and freestyle disciplines. The success of our framework validates computer vision's potential to transform traditionally manual snow management practices into automated, data-driven processes.

References

[1] Dimitris Bertsimas, David B Brown, and Constantine Caramanis. Theory and applications of robust optimization. *SIAM review*, 53(3):464–501, 2011. 2

[2] John R Birge and François Louveaux. Introduction to stochastic programming. 2011. 2

[3] Roberto Buizza. Introduction to the special issue on "25 years of ensemble forecasting". *Quarterly Journal of the Royal Meteorological Society*, 144(S1):3–6, 2018. 2

[4] Pirmin Philipp Ebner, Franziska Koch, Valentina Premier, Carlo Marin, Florian Hanzer, Carlo Maria Carmagnola, Hugues François, Daniel Günther, Fabiano Monti, Olivier Hargoaa, et al. Evaluating a prediction system for snow management. *The Cryosphere Discussions*, 2021:1–39, 2021. 5

[5] Yuan Gao, Huan Lv, Jie Zhang, Yong Liu, and Changjian Wang. Snow property extraction from visual data: A review and dataset. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3520–3529, 2021. 2

[6] Florian Hanzer, Carlo Maria Carmagnola, Pirmin Philipp Ebner, Franziska Koch, Fabiano Monti, Mathias Bavay, Matthias Bernhardt, Matthieu Lafaysse, Michael Lehning, Ulrich Strasser, et al. Simulation of snow management in alpine ski resorts using three different snow models. *Cold Regions Science and Technology*, 172:102995, 2020. 1

[7] Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, et al. The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. 2

[8] Regine Hock. Temperature index melt modelling in mountain areas. *Journal of Hydrology*, 282(1-4):104–115, 2003. 2

[9] Judith Koeberl, Hugues Francois, Jonathan Cognard, Carlo Carmagnola, Franz Prettenthaler, Andrea Damm, and Samuel Morin. The demand side of climate services for real-time snow management in alpine ski resorts: Some empirical insights and implications for climate services development. *Climate Services*, 22:100238, 2021. 2

[10] Rong Liu, Jie Shen, Honghai Wang, Chaoyang Chen, and Yangang Wang. Monocular 3d human pose estimation for sports analysis. *IEEE Transactions on Multimedia*, 24:2135–2147, 2021. 2

[11] Matt Nolan, Christopher Larsen, and Matthew Sturm. Mapping snow depth from manned aircraft on landscape scales at centimeter resolution using structure-from-motion photogrammetry. *The Cryosphere*, 9(4):1445–1463, 2015. 2

[12] Yu Peng, Yong Liu, Zheng Li, and Changjian Wang. Energy consumption and greenhouse gas emissions of artificial snowmaking. *Journal of Cleaner Production*, 382:135301, 2023. 2

[13] Karl Rittger, Kat J Bormann, and Thomas H Painter. Can machine learning help us predict snow cover**?**. *Frontiers in Earth Science*, 8:10, 2020. 2

[14] Tobias Ruppert, Michael Fassl, Rainer Stiefelhagen, and Marcus Hutter. Adas: A large-scale aerial dataset for semantic segmentation in alpine environments. *International Journal of Computer Vision*, 130(10):2513–2532, 2022. 2

[15] Pierre Spandre, Hugues François, Emmanuelle George-Marcelpoil, and Samuel Morin. Panel based assessment of snow management operations in french ski resorts. *Journal of Outdoor Recreation and Tourism*, 16:24–36, 2016. 2

[16] Robert Steiger, Daniel Scott, Bruno Abegg, Marc Pons, and Carlo Aall. Ski tourism in a warmer world: Increased adaptation and regional economic impacts in austria. *Tourism Management*, 77:104032, 2019. 1, 2

[17] Xi Wang and Duan Li. Model predictive control for supply chain management. *IEEE Transactions on Control Systems Technology*, 21(3):822–835, 2012. 2

[18] Liu Yang, Mingyang Gao, and Wenfeng Xu. Automatic real-time data collection method for ice and snow sports based on deep learning. *International Journal of High Speed Electronics and Systems*, page 2540310, 2025. 1

[19] Di Zhang and Suvrajeet Sen. A stochastic conjugate subgradient algorithm for two-stage stochastic programming. *arXiv preprint arXiv:2503.21053*, 2025. 2